QQ-P-00428c(DSA-DM)
28 February 1967
Interim Revision of
Fed. Spec. QQ-P-428b
26 February 1957

INTERIM FEDERAL SPECIFICATION

PLATINUM FOIL (DENTAL)

This interim Federal specification was developed by the Defense Personnel Support Center, 2800 South 20th Street, Philadelphia, Pennsylvania 19101, based upon currently available technical information. It is recommended that Federal agencies use it in procurement and forward recommendations for changes to the preparing activity at the address shown above. The General Services Administration has authorized the use of this interim Federal specification as a valid exception to Federal Specification QQ-P-428b.

1. SCOPE

1.1 Scope. This specification covers one type of platinum foil suitable for dental use.

2. APPLICABLE DOCUMENTS

2.1 Specifications and standards. The following specifications and standards, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein:

SPECIFICATIONS

Federal

PPP-B-566	Boxes, Folding, Paperboard.
PPP-B-585	Boxes, Wood, Wirebound.
PPP-B-601	Boxes, Wood, Cleated-Plywood.
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard.
PPP-B-676	Box, Setup.

Military

MIL-L-10547	Liners, Case, and Sheet, Overwrap; Water-Vaporproof, or Waterproof, Flexible.

FSC 6520

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STANDARDS

Federal

FED-STD-123 Marking for Domestic Shipment (Civilian Agencies).

Military

MIL-STD-105 Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-129 Marking for Shipment and Storage.

(Activities outside the Federal Government may obtain copies of Federal Specifications and Standards as outlined under General Information in the Index of Federal Specifications and Standards and at the prices indicated in the Index. The Index, which includes cumulative monthly supplements as issued, is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, D. C. 20402.

(Single copies of this specification and other product specifications required by activities outside the Federal Government for bidding purposes are available without charge at the General Services Administration Regional Offices in Boston, New York, Atlanta, Chicago, Kansas City, Mo., Dallas, Denver, San Francisco, Los Angeles, Seattle, and Washington, D. C.

(Federal Government activities may obtain copies of Federal Specifications and Standards and the Index of Federal Specifications and Standards from established distribution points in their agencies).

(Copies of Military specifications and standards required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publication. The following document forms a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

U. S. Department of Commerce
National Bureau of Standards

Bulletin of the Bureau of Standards, No. 254, 1915.

(Application for copies should be addressed to the U. S. Department of Commerce, National Bureau of Standards, Washington, D. C. 20234.)

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3. REQUIREMENTS

3.1 Material. Foil shall be a minimum of 99.9 percent pure platinum. Foil shall have a thermoelectric potential not greater than 0.15 millivolt, when tested as specified in 4.4.1.

3.2 Design and construction. The platinum foil shall be suitable in all respects for dental inlay or porcelain jacket matrices. Foil shall be furnished in 1-pennyweight, wrinkle-free sheets, each 1-1/4 ± 1/8 inches wide and 3-5/8 ± 1/8 inches long. Average thickness of the foil shall measure 0.0010 ± 0.0001 inch, when determined as specified in 4.4.2.

3.3 Weight tolerance. Ten randomly selected sheets of foil from each lot shall show no minus tolerance, when weighed to the nearest 0.01 gram.

3.4 Softness. The foil shall be sufficiently soft to comply with the test requirements specified in 4.4.3. At completion of the test, the distance between the ends of the foil shall be not greater than 20 millimeters (mm.).

3.5 Lot or control number. Each unit package shall be marked with the lot or control number.

3.6 Workmanship. Workmanship shall be first class throughout. The platinum foil shall be free from defects which detract from its appearance or may impair its serviceability.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification, where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.1.1 Inspection. Inspection, as used in this specification, is defined as both examination (such as visual or auditory investigation without the use of special laboratory appliances or procedures) and testing (determination by technical means of physical and chemical properties).

4.1.2 Certificates of quality. When available, certificates of quality, supplied by the manufacturer of the component or material, listing the specified test method and test results obtained, may be furnished in lieu of actual performance of such testing by the contractor.

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4.2 Sampling.

4.2.1 For examination. Sampling for examination shall be conducted in accordance with MIL-STD-105.

4.2.2 For tests. Sampling for tests shall be conducted in accordance with Table I. The number of determinations per lot or batch number shall be as specified in Table I. Failure in any determination shall be cause for rejection of the lot or batch.

Table I. Test Criteria

Characteristic	Reqmt.	Test procedure	Unit(s) of product to be tested per determination	Number of determinations
Purity	3.1	4.4.1	1	2
Thickness	3.2	4.4.2	10*	3
Softness	3.4	4.4.3	1	5

* 10 sheets of foil are required for each determination.

4.3 Examination. The platinum foil shall be examined to determine compliance with specification requirements.

4.4 Tests.

4.4.1 Purity. The purity of two specimens from each lot shall be determined by a thermoelectric test against pure platinum, using the method of Burgess and Sale (see 6.3), or other suitable methods of equal accuracy. The test shall be conducted with a difference of 1200 degrees Centigrade (°C.) between the hot and cold junctions.

4.4.2 Thickness. The thickness of the foil shall be determined by measuring a layer of ten 1-pennyweight sheets from each lot with a ratchet micrometer. The average thickness of the ten sheets shall be regarded as the individual thickness.

4.4.3 Softness. A bend test for softness shall be made on a minimum of five specimens of foil per lot. Each specimen shall be cut to approximately 63 millimeters (mm.) in length and folded around a steel rod 20 mm. in diameter, so that the ends will meet. The end of the rod shall be placed against a smooth glass plate, and the foil permitted to spring open as it stands with the rod in a vertical position on the glass plate. Distance between the ends of the foil shall be measured, and reported individually.

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5. PREPARATION FOR DELIVERY

5.1 Packaging.

5.1.1 Unit of issue. One (1) sheet, as specified, constitutes one unit of issue.

5.1.2 Packaging quantities. The number of units of issue indicated in table II shall be packaged in each unit, intermediate, and exterior container, as applicable, for the required level of protection specified in the procurement document.

Table II. Packaging Quantities

Unit package	Intermediate package	Exterior container
1 unit	20 units	1000 units

5.1.2.1 Packing variation permitted. If the required number of units to be shipped is less than the number of units specified to be overpacked in an exterior container, such units shall be packed in an exterior container of suitable size and design, acceptable to a common carrier, which shall insure safe delivery to destination.

5.1.3 Level A.

5.1.3.1 Unit package. Each unit shall be backed on both sides with chipboard of adequate weight to prevent bending, and then shall be packaged in an envelope of suitable size and design. Envelope shall be adequately secured to prevent accidental opening.

5.1.3.2 Intermediate package. Intermediate package shall be a carton of appropriate size constructed in accordance with PPP-B-566 or PPP-B-676, except that commercial colors will be acceptable. Closure shall be adequate to prevent spilling of contents under normal handling.

5.1.4 Level C. Units shall be packaged in standard commercial containers of the size and kind commonly used, which will afford the degree of protection required for shipment and use of the product for its intended purpose.

5.2 Packing.

5.2.1 Level A.

5.2.1.1 Exterior container. Exterior container shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 3 use; PPP-B-601, table II, using type III, class 1 plywood; PPP-B-621, table III; or PPP-B-636, type CF, class weather-resistant.

5.2.1.2 Case liner. Each level A wood box shall be lined with a waterproof case liner conforming to MIL-L-10547. Closure and sealing shall conform to applicable paragraphs of appendix thereto. Case liner shall not be required for fiberboard boxes.

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5.2.1.3 Closure. Closure of wood boxes shall be in accordance with appendix of applicable box specification. Closure of each fiberboard box shall be as specified in the appendix, and waterproofing shall conform with 30.4 of PPP-B-636.

5.2.1.4 Strapping. Strapping, when required, shall be in accordance with appendix of applicable box specification.

NOTE: Strapping shall not be required for shipments forwarded to a receiving activity within the continental limits of the United States for storage and redistribution.

5.2.2 Level B.

5.2.2.1 Exterior container. Exterior container shall conform to PPP-B-601, table I, using type III, class 1 plywood; or shall be designed for type 2 load, and constructed in accordance with PPP-B-585, style 3, for class 2 use; PPP-B-621, table I; or PPP-B-636, type CF, class domestic and method I closure shall be utilized.

5.2.3 Level C. The subject commodity shall be packed in substantial commercial containers of the type, size, and kind commonly used for the purpose, so constructed as to insure acceptance and safe delivery by common or other carriers, at the lowest rate, to point of delivery called for in the contract or purchase order.

5.3 Marking.

5.3.1 Civilian agencies. Marking of shipments shall be in accordance with FED-STD-123.

5.3.2 Military.

5.3.2.1 Interior packages. Each unit package and each intermediate package shall be marked in accordance with MIL-STD-129. When labels are utilized, waterproofing shall be required only when applicable carton is fabricated of water-resistant material. Contract or purchase order number and name of contractor shall be shown.

5.3.2.2 Exterior container. Exterior container shall be marked in accordance with MIL-STD-129.

NOTE: No reference to item description or to the word "Platinum" shall appear on the shipping containers.

6. NOTES

6.1 Ordering data. Procurement documents should specify the following:

(a) Number and date of this specification.
(b) Selection of applicable levels of packaging and packing (see 5.1 and 5.2).

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6.2 This specification does not cover all types, classes, grades, or sizes of the commodity indicated by the title of this specification, or those which are commercially available, but is intended to cover the types which are normally procured to meet requirements.

6.3 Purity determination. Purity determination by the method of George K. Burgess and P.D. Sale may be found in the Bulletin of the Bureau of Standards, Number 254, 1915, "A Study of the Quality of Platinum Ware," pages 289 to 316 (see 4.4.1).

Custodians:
Army - MD
Navy - MS
Air Force - 03

Preparing activity:
Defense Supply Agency - DM

Review activities:
Army - MD
Navy - MS
Air Force - 03

Review information is current as of the date of this document. For future coordination of changes to this document, draft circulation should be based on the information in the current DODISS.

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CAUTION

NOTICE TO BIDDERS/OFFERORS

DO NOT CONDITION OR BASE YOUR BID/OFFER ON ANY CURRENT PROCUREMENT ON THE INFORMATION SUBMITTED ON THIS FORM SINCE ANY CHANGES OR DELETIONS IN THE SPECIFICATIONS MAY RENDER YOUR BID/OFFER NON-RESPONSIVE IN WHICH CASE IT CANNOT BE CONSIDERED FOR AWARD.

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION
QQ-P-00428c(DSA-DM)

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

DD FORM 1 JAN 66 1426 REPLACES EDITION OF 1 OCT 64, WHICH MAY BE USED.

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